Adam Torres Encarnacion

University Park, PA | Email | LinkedIn | GitHub | Portfolio

EDUCATION

The Pennsylvania State University — ***Expected:*** May 2028
M.S. in Artificial Intelligence

The Pennsylvania State University — ***Expected:*** Dec 2026
B.S. in Data Science, B.S. in Statistics — ***GPA:*** *3.7*
Relevant Coursework: Machine Learning, Matrix Algebra, Numerical Analysis, Data Structures & Algorithms, Big Data

WORK EXPERIENCE

Amazon — May 2026 – Aug 2026
Software Engineering Intern — *Cupertino, CA*

- Refactored **C++** infrastructure for 2 fiber-cleaning robot arms projected to save **$28.6M/year** across **85** data centers
- Built a 19-route **HTTP** server to control robot arms remotely over **SSH**, maintaining the local control loop **under 2ms**
- Made an LLM-as-judge classification pipeline for 6 optical fiber and transceiver labels, storing live detection data in **S3**
- Created a self-evolving **MCP** knowledge base for agent-first development, migrating ∼**400** legacy docs from 2 repos
- Hardened 2 **AWS CI/CD** pipeline stages with blocking lint checks, code complexity checks, and ∼**40** integration tests

IBM — May 2025 – May 2026
AI Engineering Intern — *Remote*

- Shipped an LLM-powered **REST** microservice to power documentation for **3,100+** scripts, saving **$223,000** in labor costs
- Helped triage ∼**500k** criminal records spanning 20+ years with a **LangChain** agent built on **Python**, cutting review time
- Deployed serverless **Azure Function Apps** that autoscaled with incoming jobs, cutting manual review workload by **78%**
- Refined a file-chunking pipeline using **Llama-4** and **GPT-4o-mini** to handle 7+ large-scale internal processing requests

PROJECTS

Agent-Driven Vision Annotation | *Python, WebMCP, OpenCV*

- Turned AI agents into click-free CV data annotators through **26 WebMCP tools** to draw & navigate an open canvas
- Built an OpenCV vision keyboard emulator with **98.78% F1** on 1,253 AI-labeled images after 600 MPS-accelerated epochs
- Balanced image datasets using 8 deterministic image transforms and GPT-Image-2 in asynchronous augmentation jobs
- Supported **16 concurrent labeling sessions** at <300 ms p95 metadata API latency using FastAPI, PostgreSQL, and **S3**

Multimodal Drawing Canvas **(NexHacks '26 ByteDance 2nd place)** | *TypeScript, Gemini, PostgreSQL*

- Won 2nd (out of 1500+) **solo** with a tldraw canvas that auto-completes sketches in place via Gemini AI image generation
- Used Gemini to **classify** chat vs. drawing requests, then cropped subjects with **Grounding DINO** on **CUDA** (PyTorch)
- Integrated OpenAI Whisper for voice prompts and gated each image generation behind an on-canvas accept/reject overlay
- Saved tldraw snapshots as **PostgreSQL JSONB** on a 2s debounce, generating PNG previews and skipping saves offline

Multi-Agent Social Simulation **(YHack '26 K2 Think V2 1st place)** | *Python, LangGraph, asyncio*

- Built a multi-agent simulation engine with FastAPI and **LangGraph StateGraph**, using asyncio for concurrent agent execution, shared memory streams, and Pydantic validation for structured outputs
- Modeled echo-chamber formation across **50+ agents** using recency-weighted memory retrieval, threshold-triggered reflection, and Deffuant bounded-confidence opinion updates; streamed simulation updates over Socket.IO

AWARDS ($15K+)

- **1st Place** at c0mpiled 15, AWS Track — Oct 2026
- **1st Place** at HopHacks, Tiger Data Track — Sep 2026
- **1st Place** at YHack, K2 Think V2 Track — Mar 2026
- **2nd Place** at NexHacks, ByteDance Track (Solo) — Jan 2026
- **1st Place** at HackPrinceton, xAI Track — Nov 2025
- Awarded a Jane Street **$5,000** scholarship (AMP), 1 of 80 recipients nationally — Jul 2023

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, TypeScript, HTML/CSS, R, Lua
Frameworks: PyTorch, OpenCV, Hugging Face, Pandas, NumPy, FastAPI, React, Node.js, Next.js, LangChain/LangGraph
Developer Tools: Git, Linux, VS Code, Vim/Neovim, Pi (harness), Cmux (terminal multiplexer), Agile/Scrum
Databases: PostgreSQL, MySQL, SQLite, Redis, MongoDB, DynamoDB
Cloud/Infra: AWS (Lambda, S3, EC2, Bedrock, DynamoDB), Azure Functions, Vercel, Docker, CI/CD (GitHub Actions)